Lecture Notes:
Commutative Algebra
&
Introduction to Algebraic Geometry

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(Based on the notes taken by Friedrich Günther and Niklas Müller in Summer Term 2019)

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Chapter 1

Rings and Ideals

In this chapter, we introduce and study some basic notions and results concerning rings and ideals. Note that some of them were already presented in the courses "Lineare Algebra I" and "Algebra", but are included here for completeness and clarity.

Throughout these notes, all rings are assumed to be *commutative with identity*, unless explicitly stated otherwise. In particular, any nontrivial ring $R \neq \{0\}$ contains at least two different elements: 0 and 1.

Definition 1.1. Let R and S be rings. A *ring homomorphism* $\varphi\colon R \to S$ is a map fulfilling the following properties:

(a) $\varphi(x+y) = \varphi(x) + \varphi(y)$ for all $x, y \in R$,

(b) $\varphi(xy) = \varphi(x)\varphi(y)$ for all $x, y \in R$,

(c) $\varphi(1_R) = 1_S$.

Note that property (c) does not hold automatically and hence must be required.

Example 1.2. Let R and S be nontrivial rings. The following maps between rings fulfill (a) and (b), but not (c).

(1) The zero map $R \to S,\ x \mapsto 0$.

(2) The map $R \to R \times R,\ x \mapsto (x, 0)$.

(3) The map $\mathbb{Z}/6\mathbb{Z} \to \mathbb{Z}/6\mathbb{Z},\ \overline{n} \mapsto \overline{3} \cdot \overline{n}$.

Definition 1.3. Let R be a ring. A subset $S \subseteq R$ is a *subring* of R if it is a ring with respect to the operations induced by those in R and moreover $1_R \in S$.

Remark 1.4. Let R be a ring and let $S \subseteq R$ be a subset.

(a) The subset S is a subring if and only if it is nonempty, closed under subtraction and under multiplication, and $1_R \in S$.

(b) If S is a subring, then the inclusion map $S \to R$, $x \mapsto x$, is a ring homomorphism.

Remark 1.5. In the definition of a subring $S \subseteq R$, the condition $1_R \in S$ is not automatically satisfied and must be imposed. Indeed, a subset of a ring may itself form a ring under the induced operations without containing the identity element of the larger ring. For example, let R be a nontrivial ring and consider

$$S := \{(x, 0) \mid x \in R\} \subseteq R \times R.$$

The subset S is a ring under the operations inherited from $R \times R$: indeed, S is nonempty, closed under subtraction and multiplication, and $(1, 0)$ is the multiplicative identity of S. However, the identity element of $R \times R$ is $(1, 1)$ and since R is nontrivial we have $(1, 0) \neq (1, 1)$. Since $(1, 1) \notin S$, the ring S is not a subring of $R \times R$.

Definition 1.6. Let R be a ring and let $a, b \in R$. We say that b *divides* a, or that b *is a divisor of* a, and write $b \mid a$, if there exists an element $c \in R$ such that $a = bc$.

Definition 1.7. Let R be a ring and let $r \in R$.

(a) The element r is called a *unit* if it is invertible with respect to multiplication, i.e., there exists $s \in R$ such that $rs = 1$. We denote by R^* the set of all units in R. Notice that $(R^*, \cdot)$ is an abelian group.

(b) The element r is called a *zero divisor* if there exists $s \in R \setminus \{0\}$ such that $rs = 0$. A nontrivial ring whose only zero divisor is 0 is called an *integral domain*.

(c) The element r is called *nilpotent* if there exists $n \in \mathbb{N}_{>0}$ such that $r^n = 0$.

(d) Assume now that $r \neq 0$ and $r \notin R^*$.

(i) The element r is called *prime* if for all $a, b \in R$

$$r \mid ab \quad \Longrightarrow \quad r \mid a \quad \text{or} \quad r \mid b.$$

(ii) The element r is called *irreducible* if for all $a, b \in R$

$$r = ab \quad \Longrightarrow \quad a \in R^* \quad \text{or} \quad b \in R^*.$$

Remark 1.8.

(a) In a nontrivial ring any nilpotent element is a zero divisor. However, the converse is not true in general. For instance, in the ring $\mathbb{Z}/6\mathbb{Z}$ the element $\overline{2}$ is a zero divisor, but it is not nilpotent.

(b) In general, the notions of prime and irreducible elements are independent: a prime element need not be irreducible, and an irreducible element need not be prime. For instance, in the ring $\mathbb{Z} \times \mathbb{Z}$ the element $(1,0)$ is prime, but not irreducible since $(1,0) = (1,0) \cdot (1,0)$. Also, in the ring $\mathbb{Z}[i\sqrt{5}] := \{a + bi\sqrt{5} \mid a, b \in \mathbb{Z}\}$ the element 2 is irreducible, but not prime. Thus, without additional assumptions on the ring, there is no implication between the two concepts. However, in special classes of rings the situation improves:

(i) In an integral domain, every prime element is irreducible.

(ii) In a unique factorization domain (UFD), the two notions coincide.

1.1 Ideals: Examples and Operations

Definition 1.9. Let $(R, +, \cdot)$ be a ring. A subset $I \subseteq R$ is called an *ideal* if:

(a) $(I, +)$ is a subgroup of $(R, +)$, and

(b) for all $r \in R$, for all $a \in I$, it holds $ra \in I$.

We give now some examples of ideals and define operations on them.

Example 1.10. Let R be a ring.

(a) The subsets $\{0\}$ and R are ideal and are called *trivial ideals*.

(b) Given two ideals $I, J \subseteq R$, we define

$$\begin{aligned} I + J &:= \Big\{ i + j \mid i \in I,\ j \in J \Big\}, \\ IJ &:= \left\{ \sum_{k=1}^{n} i_k j_k \mid n \in \mathbb{N},\ i_k \in I,\ j_k \in J \right\}, \\ I \cap J &:= \Big\{ a \mid a \in I \text{ and } a \in J \Big\}. \end{aligned}$$

All three of them are ideals and it holds $IJ \subseteq I \cap J$. Notice that

$$I \cup J := \Big\{ a \mid a \in I \text{ or } a \in J \Big\}$$

is an ideal if and only if $I \subseteq J$ or $J \subseteq I$.

(c) Let $\{I_\alpha\}_{\alpha \in \mathcal{A}}$ be a family of ideals in R. We define

$$\begin{aligned} \sum_{\alpha \in \mathcal{A}} I_\alpha &:= \left\{ \sum_{\alpha \in \mathcal{A}} i_\alpha \;\middle|\; i_\alpha \in I_\alpha \text{ for all } \alpha \in \mathcal{A} \text{ and } i_\alpha \neq 0 \text{ only for finitely many } \alpha \right\}, \\ \bigcap_{\alpha \in \mathcal{A}} I_\alpha &:= \Big\{ a \;\Big|\; a \in I_\alpha \text{ for all } \alpha \in \mathcal{A} \Big\}. \end{aligned}$$

Both of them are ideals. Now, assume that $\{I_\alpha\}_{\alpha\in\mathcal{A}}$ is a chain of ideals, i.e., for any $\alpha, \beta \in \mathcal{A}$, either $I_\alpha \subseteq I_\beta$ or $I_\beta \subseteq I_\alpha$. Then,

$$\bigcup_{\alpha\in\mathcal{A}} I_\alpha := \Big\{a \;\Big|\; a \in I_\alpha \text{ for some } \alpha \in \mathcal{A}\Big\}$$

is also an ideal.

(d) Let $a \in R$. We define

$$(a) := \{\lambda a \mid \lambda \in R\}.$$

This is an ideal called *the ideal generated by* a; we denote it by aR as well.

(e) Let $I \subseteq R$ be an ideal. We say that I is *principal* if there exists $a \in R$ such that $I = (a)$.

(f) Let $a_1, \ldots, a_n \in R$. We define

$$(a_1, \ldots, a_n) := a_1R + \ldots + a_nR = \left\{ \sum_{i=1}^{n} \lambda_i a_i \;\Big|\; \lambda_i \in R \right\}.$$

This is called *the ideal generated by* $a_1, \ldots, a_n$.

(g) Let $\{a_i\}_{i\in\mathcal{A}}$ be a subset of R. We define

$$(a_i)_{i\in\mathcal{A}} := \sum_{i\in\mathcal{A}} a_i R_i.$$

This is called *the ideal generated by the subset* $\{a_i\}_{i\in\mathcal{A}}$.

(h) Let $I \subseteq R$ be an ideal. We say that the subset $\{a_i \mid i \in \mathcal{A}\} \subseteq I$ is a *system of generators for* I if $I = (a_i)_{i\in\mathcal{A}}$. The ideal I is said to be *finitely generated* if there exists a finite system of generators for I.

Remark 1.11.

(a) Let R be a ring and let $I \subseteq R$ be an ideal. If $1_R \in I$, then $I = R$. In particular, a proper ideal $I \subsetneq R$ is not a subring of R.

(b) A proper ideal $I \subsetneq R$ of a ring R might be itself a ring. In this case, $1_I \neq 1_R$ is an idempotent element of R. For instance, the ring $S \subseteq R \times R$ given in Remark 1.5 is an ideal in $R \times R$. A natural question then arises: does there exist an example of a subset of a ring that is itself a ring but neither a subring nor an ideal of the ambient ring? The answer is positive. Indeed, the same ring S can be regarded as a subset of the polynomial ring $(R \times R)[x]$, but it is neither a subring nor an ideal of $(R \times R)[x]$.

1.2 Prime Ideals and Maximal Ideals

Definition 1.12. Let R be a ring and let $I \subsetneq R$ be a proper ideal.

(a) The ideal I is called *prime* if for any $a, b \in I$

$$ab \in I \implies a \in I \text{ or } b \in I.$$

(b) The ideal I is called *maximal* if for any ideal J

$$I \subseteq J \subseteq R \implies J = I \text{ or } J = R.$$

Proposition 1.13. *Let R be a ring and let $I \subsetneq R$ be a proper ideal.*

(i) The ideal I is prime if and only if R/I is an integral domain.

(ii) The ideal I is maximal if and only if R/I is a field.

Corollary 1.14. *In a ring any maximal ideal is prime.*

The converse does not hold true in general.

Example 1.15. Let R be a ring and let $I \subsetneq R$ be a proper ideal.

(a) If R is an integral domain that is not a field, then $I = (0)$ is prime, but not maximal.

(b) In $R = \mathbb{Z}[x, y]$, the ideal $I = (x)$ is prime, but not maximal since

$$(x) \subsetneq (x, y) \subsetneq \mathbb{Z}[x, y].$$

Proposition 1.16. *Let R be a ring and let $I \subseteq R$ be an ideal. There is a one-to-one inclusion-preserving correspondence between the ideals of R containing I and the ideals of R/I. Under this correspondence, prime (resp. maximal) ideals correspond to prime (resp. maximal) ideals.*

Hint: Let $\pi \colon R \to R/I$ be the canonical projection. If J is an ideal in R/I, then $\pi^{-1}(J)$ is an ideal in R containing I.

Proof. Exercise. □

Proposition 1.17. *(i) Every nontrivial ring $R \neq 0$ has a maximal ideal.*

(ii) Let R be a ring and let $I \subsetneq R$ be a proper ideal. Then there exists in R a maximal ideal containing I.

Proof. (i) Let $\mathcal{I} := \{I' \subsetneq R \mid I' \text{ is an ideal of } R\}$. Since $(0) \in \mathcal{I}$, the set $\mathcal{I}$ ordered by inclusion is a nonempty partially ordered set (poset for short). Let $\{I_\alpha\}_{\alpha\in\mathcal{A}}$ be a chain in $\mathcal{I}$, i.e., a totally ordered family of proper ideals in R. Since $\bigcup_{\alpha\in\mathcal{A}} I_\alpha$ is a proper ideal, every totally ordered subset of $\mathcal{I}$ has an upper bound in $\mathcal{I}$. Hence, by Zorn's Lemma, there exists a maximal element for the poset $(\mathcal{I}, \subseteq)$, i.e., a maximal ideal in R.

(ii) Apply (i) to the quotient ring R/I and then the one-to-one inclusion-preserving correspondence given by Proposition 1.16. Alternatively, argue as in (i) setting

$$\mathcal{I} := \{I \subseteq I' \subsetneq R \mid I' \text{ is an ideal of } R\}.$$

□

Corollary 1.18. *Every non-unit of a ring R is contained in a maximal ideal.*

Proof. Let $r \in R \setminus R^*$. Then the principal ideal (r) is a proper ideal of R. Now we apply Proposition 1.17 (ii) and we are done. □

Proposition 1.19. *Let R be a ring.*

(i) Let $\mathfrak{p}_1, \ldots, \mathfrak{p}_n \subsetneq R$ be prime ideals and let $I \subseteq R$ be an ideal such that $I \subseteq \bigcup_{i=1}^n \mathfrak{p}_i$. Then $I \subseteq \mathfrak{p}_i$ for some $i \in \{1, \ldots, n\}$.

(ii) Let $I_1, \ldots, I_n \subseteq R$ be ideals and let $\mathfrak{p} \subsetneq R$ be a prime ideal such that $\mathfrak{p} \supseteq \bigcap_{j=1}^n I_j$. Then $\mathfrak{p} \supseteq I_j$ for some $j \in \{1, \ldots, n\}$. Moreover, if $\mathfrak{p} = \bigcap_{j=1}^n I_j$, then $\mathfrak{p} = I_j$ for some $j \in \{1, \ldots, n\}$.

Proof. (i) The statement will be shown by induction on n in the form: If $I \not\subseteq \mathfrak{p}_i$ for $1 \le i \le n$, then $I \not\subseteq \bigcup_{i=1}^n \mathfrak{p}$.

For $n = 1$ the statement is trivial. Let now $n \ge 2$ and assume the statement is true for $n - 1$. Then for each $1 \le i \le n$ by the inductive hypothesis there exists $x_i \in I$ such that $x_i \notin \mathfrak{p}_j$ for all $j \ne i$. If for some $1 \le i \le n$ it holds $x_i \notin \mathfrak{p}_i$, then $x_i \notin \mathfrak{p}_j$ for all $1 \le j \le n$, thus we are done. Otherwise, it holds $x_i \in \mathfrak{p}_i$ for all $1 \le i \le n$. Consider the element

$$y := \sum_{i=1}^n x_1 x_2 \ldots x_{i-1} x_{i+1} \ldots x_n.$$

Note that $y \in I$ and $y \notin \mathfrak{p}_i$ for all $1 \le i \le n$. Indeed, assume by contradiction that $y \in \mathfrak{p}_i$ for some i. Then $x_1 x_2 \ldots x_{i-1} x_{i+1} \ldots x_n \in \mathfrak{p}_i$ since each of the other summands is a multiple of $x_i \in \mathfrak{p}_i$ and hence in $\mathfrak{p}_i$. But then, since $\mathfrak{p}_i$ is prime, we would have $x_j \in \mathfrak{p}_i$ for some $j \ne i$, a contradiction. Hence $y \notin \bigcup_{i=1}^n \mathfrak{p}_i$ and we are done.

(ii) Assume that $\mathfrak{p} \not\supseteq I_j$ for all $1 \leq j \leq n$. Then there exists $x_j \in I_j$ but $x_j \notin \mathfrak{p}$ for each $1 \leq j \leq n$. Thus

$$\prod_{j=1}^{n} x_j \in \prod_{j=1}^{n} I_j \subseteq \bigcap_{j=1}^{n} I_j.$$

But, since $\mathfrak{p}$ is prime, $\prod_{j=1}^{n} x_j \notin \mathfrak{p}$ and thus $\mathfrak{p} \not\supseteq \bigcap_{j=1}^{n} I_j$.

Now, if $\mathfrak{p} = \bigcap_{j=1}^{n} I_j$, then in particular we have $\mathfrak{p} \supseteq \bigcap_{j=1}^{n} I_j$ and therefore there is some $j_0 \in \{1, \ldots, n\}$ such that $I_{j_0} \subseteq \mathfrak{p}$. On the other hand, $\mathfrak{p} \subseteq I_j$ for $1 \leq j \leq n$, hence $\mathfrak{p} = I_{j_0}$.

□

1.3 Nilradical, Jacobson Radical and Local Rings

Proposition/Definition 1.20. *Let R be a ring and let $I \subseteq R$ be an ideal. The set*

$$\sqrt{I} := \{a \in R \mid a^n \in I \text{ for some } n \in \mathbb{N}_{>0}\}$$

is an ideal called the radical of I.

Proof. Let $r \in R$ and let $a \in \sqrt{I}$. Then $a^n \in I$ for some $n \in \mathbb{N}_{>0}$ and so $(ra)^n = r^n a^n \in I$ since I is an ideal. Now let $x, y \in \sqrt{I}$. Then there exist $n, m \in \mathbb{N}_{>0}$ such that $x^n, y^m \in I$. By the binomial formula we get

$$(x+y)^{m+n-1} = \sum_{k=0}^{m+n-1} \binom{m+n-1}{k} \underbrace{x^k y^{m+n-1-k}}_{\in I} \quad ,$$

hence, $(x+y)^{m+n-1} \in I$, which implies $x + y \in \sqrt{I}$. Thus, we have shown that $\sqrt{I}$ is indeed an ideal.

□

Proposition 1.21. *Let R be a ring. Let I and J be two ideals in R. Then:*

(i) $\sqrt{I} \supseteq I$,

(ii) $\sqrt{\sqrt{I}} = \sqrt{I}$,

(iii) $\sqrt{IJ} = \sqrt{I \cap J} = \sqrt{I} \cap \sqrt{J}$.

(iv) $\sqrt{I} = R \iff I = R$.

Definition 1.22. Let R be a ring. The *nilradical of R* is defined as

$$N_R := \sqrt{(0)} = \{a \in R \mid a^n = 0 \text{ for some } n \in \mathbb{N}_{>0}\}.$$

In other words, the set of nilpotent elements in R is indeed an ideal called the nilradical of R.

Proposition 1.23. *Let R be a ring. Then in the quotient ring R/N_R the only nilpotent element is* 0.

Proof. Let $[x] \in R/N_R$ be nilpotent. This means there is $k \in \mathbb{N}$ such that $[x]^k = [x^k] = 0$ in R/N_R, hence $x^k \in N_R$. Because $x^k \in N_R$, there is $l \in \mathbb{N}$ such that $(x^k)^l = x^{kl} = 0$ in R, and thus $x \in N_R$. Because $x \in N_R$, $[x] = 0$ in R/N_R. □

The following theorem describes the nilradical of a ring and, more in general, the radical of any ideal.

Theorem 1.24. *Let R be a ring and let $I \subseteq R$ be an ideal. Then:*

(i) The nilradical of R is the intersection of all prime ideals of R, i.e.,

$$N_R = \bigcap_{\substack{P \subsetneq R \\ prime}} P.$$

(ii) The radical of I is the intersection of all prime ideals containing I, i.e.,

$$\sqrt{I} = \bigcap_{\substack{I \subseteq P \subsetneq R \\ prime}} P.$$

Proof. (i) "$\subseteq$": Let $P' \subsetneq R$ be a prime ideal in R. If $x \in N_R$, there is $k \in \mathbb{N}$ such that $x^k = 0 \in P'$. Since P' is prime, it holds $x \in P'$, and thus $N \subseteq \bigcap_{\substack{P \subsetneq R \\ \text{prime}}} P$.

"$\supseteq$": Let now $x \in R \setminus N_R$ and let $\mathcal{S}_x$ be the set of proper ideals $I \subsetneq R$ with the property that if $n \in \mathbb{N}_{>0}$, then $x^n \notin I$. Since $(0) \in \mathcal{S}_x$, the set $\mathcal{S}_x$ is a nonempty poset (with respect to inclusion). As in the proof of Proposition 1.17 (i), by Zorn's Lemma $\mathcal{S}_x$ has a maximal element, say P'.

We show now that P' is a prime ideal in R. To see this, let $a, b \in R \setminus P'$. Then the ideals $(a) + P'$ and $(b) + P'$ are strictly larger than P' (with respect to inclusion) and thus are not in $\mathcal{S}_x$. Since $(a) + P'$, $(b) + P' \notin \mathcal{S}_x$, there is $m \in \mathbb{N}_{>0}$ such that $x^m \in (a) + P'$ and there is $k \in \mathbb{N}_{>0}$ such that $x^k \in (b) + P'$. Thus $x^m x^k = x^{m+k} \in [(a) + P'][(b) + P'] \subseteq (ab) + P'$. Since $x^{m+k} \in (ab) + P'$, the ideal $(ab) + P' \notin \mathcal{S}_x$, and thus $(ab) + P' \supsetneq P'$, which implies $ab \notin P'$. Finally, we have found a prime ideal P' such that $P' \in \mathcal{S}_x$. By definition of $\mathcal{S}_x$, the ideal P' does not contain any power of x. In particular, $x \notin P'$ and hence $x \notin \bigcap_{\substack{P \subsetneq R \\ \text{prime}}} P$.

(ii) Let $\pi \colon R \to R/I$ be the canonical projection. Note that $\pi^{-1}(N_{R/I}) = \sqrt{I}$. Hence, we first apply (i) to the ring R/I and then use the correspondence between prime ideals in R containing I and prime ideals in R/I given by Proposition 1.16.

□

Definition 1.25. Let R be a ring. The *Jacobson radical* J_R *of* R is definded as the intersection of all maximal ideals, i.e.,

$$J_R := \bigcap_{\substack{\mathfrak{m} \subsetneq R \\ \text{maximal}}} \mathfrak{m}.$$

Proposition 1.26. *Let R be a ring. The Jacobson radical J_R admits the following description*

$$J_R = \{x \in R \mid 1 - xy \in R^* \text{ for all } y \in R\}.$$

Proof. "$\subseteq$": Let $x \in J_R$ and assume that $1 - xy$ is not a unit for some $y \in R$. Then by Proposition 1.17 (ii) the element $1 - xy$ is contained in some maximal ideal $\mathfrak{m}$ of R and by definition of Jacobson radical, $x \in \mathfrak{m}$ as well. But then $1 \in \mathfrak{m}$, which is a contradiction.

"$\supseteq$": If $x \notin J_R$, then $x \notin \mathfrak{m}$ for some maximal ideal $\mathfrak{m}$ in R. Hence, $(x)+\mathfrak{m} \supsetneq \mathfrak{m}$ and thus $(x)+\mathfrak{m} = (1)$, i.e., there are elements $y \in R$ and $z \in \mathfrak{m}$ such that $xy+z = 1$. In other words, the element $1 - xy = z \in \mathfrak{m}$ and therefore $1 - xy$ is not a unit. □

Definition 1.27. A ring R is called *local* if it contains only one maximal ideal $\mathfrak{m}$. Then the quotient $R/\mathfrak{m}$ is a field, called the *residue field* of R. In literature, a local ring is often denoted by $(R, \mathfrak{m})$ or $(R, \mathfrak{m}, k)$, where $k := R/\mathfrak{m}$.

Proposition 1.28. *(i) Let R be a ring and let $\mathfrak{m} \subsetneq R$ be an ideal such that every element in $R \setminus \mathfrak{m}$ is a unit. Then R is a local ring and $\mathfrak{m}$ is its unique maximal ideal.*

(ii) Let R be a ring and let $\mathfrak{m} \subsetneq R$ be a maximal ideal such that every element in the set $1 + \mathfrak{m} := \{1 + x \mid x \in \mathfrak{m}\}$ is a unit. Then R is local and $\mathfrak{m}$ is its unique maximal ideal.

Proof. Exercise. □

1.4 Chinese Remainer Theorem

Definition 1.29. Let R be a ring and let $I, J \subseteq R$ be ideals. Then I and J are called *coprime* (or *comaximal*) if $I + J = R$.

Remark 1.30. Let R be a ring and I, J be ideals in R.

(i) If the ideals I, J are coprime, it holds $IJ = I \cap J$.

(ii) Two ideals $I, J \subseteq R$ are coprime if and only if there exist $i \in I$, $j \in J$ such that $i + j = 1$.

Proposition 1.31 (Chinese Remainder Theorem)**.** *Let R be a ring and $I_1, \dots, I_n \subseteq R$ be ideals. We define a ring homomorphism by*

$$\varphi\colon R \longrightarrow \prod_{j=1}^{n} (R/I_j), \qquad x \longmapsto (x + I_1, \dots, x + I_n).$$

(i) If I_k and I_l are coprime for $k \neq l$, then $\prod_{j=1}^{n} I_j = \bigcap_{j=1}^{n} I_j$.

(ii) φ is surjective if and only if I_k and I_l are coprime for $k \neq l$.

(iii) φ is injective if and only if $\bigcap_{j=1}^{n} I_j = (0)$.

1.5 Hilbert's Basis Theorem

Definition 1.32. A ring R is called *Noetherian* if every ascending chain of ideals

$$I_1 \subseteq I_2 \subseteq \dots \subseteq I_k \subseteq \dots$$

is stationary, i.e., there exists $N \in \mathbb{N}_{>0}$ such that $I_k = I_N$ for $k \geq N$.

Example 1.33. Any principal ideal domain (PID) is a Noetherian ring, as shown in the course "Algebra".

The following well-known theorem allows us to construct new Noetherian rings from a given one.

Theorem 1.34 (Hilbert's Basis Theorem)**.** *Let R be a Noetherien ring. Then $R[x_1, \dots, x_n]$ is also Noetherian.*

Proof. First of all, it suffices to show the theorem when $n = 1$ since

$$R[x_1, \dots, x_n] = R[x_1, \dots, x_{n-1}][x_n].$$

By contradiction, assume $R[x]$ is not Noetherian. Let $I_1 \subsetneq I_2 \subsetneq \dots$ be a strictly ascending chain of ideals. Set $I := \bigcup_{n \in \mathbb{N}} I_n$. We construct a sequence $\{f_1, f_2, \dots\}$ of polynomials in $R[x]$ as follows:

(a) We choose $f_1 \in I - \{0\}$ of minimal degree.

(b) Assume we found $f_1, f_2, \dots, f_{n-1}$. We pick f_n in $I \setminus (f_1, \dots, f_{n-1})$ of minimal degree.

Note that this is a well-defined construction: indeed, $\mathbb{N}$ is well-ordered and step (b) is repeated infinitely many times since the given chain of ideals is strictly ascending.

Claim 1. $\deg(f_n) \geq \deg(f_{n-1})$ for $n \geq 2$.

Proof of Claim 1. First of all, $\deg(f_2) \geq \deg(f_1)$ by (a). Assume by contradiction $\deg(f_n) < \deg(f_{n-1})$ for some $n \geq 3$. Since by construction $f_n \in I \setminus (f_1, \ldots, f_{n-1}) \subseteq I \setminus (f_1, \ldots, f_{n-2})$, we found a polynomial in $I \setminus (f_1, \ldots, f_{n-2})$ whose degree is lower than $\deg(f_{n-1})$, against the choice of f_{n-1} as a polynomial in $I \setminus (f_1, \ldots, f_{n-2})$ of minimal degree. $\square$

For $i \in \mathbb{N}$ define $a_i \in R$ to be the leading coefficient of f_i and define the ideal $J_i := (a_1, \ldots, a_i)$. We have then an ascending chain of ideals of R, namely $J_1 \subseteq J_2 \subseteq J_3 \subseteq \ldots$. As R is noetherian, there is $N \in \mathbb{N}$ such that $J_k = J_N$ for all $k \geq N$. Consider the element $a_{N+1} \in J_{N+1} = J_N = (a_1, \ldots, a_n)$. Then for $1 \leq i \leq N$ there are $u_i \in R$ such that $a_{N+1} = \sum_{i=1}^{N} u_i a_i$. Now set

$$f'_{N+1} := u_1 x^{\deg(f_{N+1}) - \deg(f_1)} f_1 + \cdots + u_N x^{\deg(f_{N+1}) - \deg(f_N)} f_N.$$

The leading term of f'_{N+1} is $a_{N+1} x^{\deg(f_{N+1})}$ and $f'_{N+1} \in (f_1, \ldots, f_N)$. Consider the polynomial

$$g := f'_{N+1} - f_{N+1}.$$

It holds $g \in I \setminus (f_1, \ldots, f_N)$ since otherwise $f_{N+1} = f'_{N+1} - g \in (f_1, \ldots, f_N)$. Moreover, by construction $\deg(g) < \deg(f_{N+1})$ and this contradicts the choice of f_{N+1} as a polynomial in $I \setminus (f_1, \ldots, f_N)$ of minimal degree. $\square$